CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

ANTERO BORGES DAIRY

2226 HULL RD

ATWATER CA 95301

PREPARED BY:

CARDOSO AG SERVICES

MARIANN CARDOSO
1306 MAIN STREET
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Antero Borges Dairy

Physical address of dairy:

2226 Hull RD	Atwater	Merced	95301
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X056-X092-X001-XXXX

B. OPERATORS

Borges, Antero & Eulalia

Operator name: Borges, Antero & Eulalia

Telephone no.: Landline: ; Cellular: (209) 358-0680

2260 Hull RD	Atwater	CA	95301
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Borges, Antero & Eulalia

Legal owner name: Borges, Antero & Eulalia

Telephone no.: Landline: ; Cellular: (209) 358-0680

2260 Hull RD	Atwater	CA	95301
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	81	69	54	62
Number under roof	321	13	0	0	0	0
Maximum number	321	13	81	69	54	62
Average number	321	13	81	69	54	62
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 10,145.13 *tons per reporting period*

Total nitrogen from manure: 127,236.74 *lbs per reporting period*

After ammonia losses (30% loss applied): 89,065.72 *lbs per reporting period*

Total phosphorus from manure: 20,823.61 *lbs per reporting period*

Total potassium from manure: 60,209.08 *lbs per reporting period*

Total salt from manure: 154,132.20 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,370,000 *gallons*

Total nitrogen generated: 7,123.68 *lbs*

Total phosphorus generated: 1,006.41 *lbs*

Total potassium generated: 13,862.88 *lbs*

Total salt generated: 62,412.26 *lbs*

	2,370,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	2,370,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Home Ranch	28	28	3	both	X056-X092-X001-XXXX
Moran Ranch	29	29	3	manure	X056-X013-X011-XXXX
Totals for areas that were used for application	57	57	6		
Totals for areas that were not used for application					
Land application area totals	57	57	6		

B. CROPS AND HARVESTS

Home Ranch

Field name: Home Ranch

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	328.00 *ton*	Dry-weight		71.2	26,200.00	4,200.00	44,700.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.71	176.78	28.34	301.61	816.44

06/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 28 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/14/2023	174.00 *ton*	Dry-weight		73.0	30,600.00	4,600.00	56,800.00		17.40
10/26/2023	185.00 *ton*	Dry-weight		80.0	30,600.00	5,700.00	48,300.00		16.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	12.82	183.56	30.50	318.25	1,012.04

Moran Ranch

Field name: Moran Ranch

10/05/2022: Oats, hay

Crop: Oats, hay Acres planted: 29 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	90.00 *ton*	Dry-weight		10.0	26,200.00	4,200.00	44,700.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	3.10	146.36	23.46	249.70	675.93

06/17/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 29 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/14/2023	104.00 *ton*	Dry-weight		72.0	17,100.00	4,800.00	41,000.00		16.10
10/26/2023	96.00 *ton*	Dry-weight		80.0	30,600.00	5,700.00	48,300.00		16.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	6.90	74.86	17.19	146.30	537.84

NUTRIENT BUDGET

A. LAND APPLICATIONS

Home Ranch - 10/05/2022: Oats, silage-soft dough

Field name: Home Ranch

Crop: Oats, silage-soft dough — Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	162.98	31.59	176.88	0.00	120.00 *ton*
Application event totals		162.98	31.59	176.88	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.16	7.79	85.76	424.88	660,000.00 *gal*
Canal	Surface water	0.13	0.00	0.00	36.75	4,403,561.00 *gal*
Application event totals		38.29	7.79	85.76	461.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	6.68	74.85	450.63	840,000.00 *gal*
Canal	Surface water	0.11	0.00	0.00	30.50	3,654,643.00 *gal*
Application event totals		34.01	6.68	74.85	481.13	

Home Ranch - 06/20/2023: Sudangrass, silage

Field name: Home Ranch

Crop: Sudangrass, silage — Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Home Ranch - 06/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.16	0.00	0.00	44.96	5,387,127.00 *gal*
Application event totals		0.16	0.00	0.00	44.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.04	10.36	161.48	653.41	420,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	50.35	6,033,061.00 *gal*
Application event totals		88.22	10.36	161.48	703.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.32	11.10	173.01	700.09	450,000.00 *gal*
Canal	Surface water	0.20	0.00	0.00	56.64	6,787,193.00 *gal*
Application event totals		94.53	11.10	173.01	756.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	60.84	10.76	51.71	0.00	168.00 *ton*
Application event totals		60.84	10.76	51.71	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.14	0.00	0.00	38.21	4,578,895.00 *gal*
Application event totals		0.14	0.00	0.00	38.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.19	0.00	0.00	53.70	6,434,570.00 *gal*
Application event totals		0.19	0.00	0.00	53.70	

Home Ranch - 06/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	42.70	5,117,282.00 *gal*
Application event totals		0.15	0.00	0.00	42.70	

Moran Ranch - 10/05/2022: Oats, hay

Field name: Moran Ranch

Crop: Oats, hay

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	199.33	38.63	216.32	0.00	152.00 *ton*
Application event totals		199.33	38.63	216.32	0.00	

10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	34.72	4,309,702.00 *gal*
Application event totals		0.12	0.00	0.00	34.72	

Moran Ranch - 06/17/2023: Sudangrass, silage

Field name: Moran Ranch

Crop: Sudangrass, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Moran Ranch - 06/17/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	92.66	16.39	78.76	0.00	265.00 *ton*
Application event totals		92.66	16.39	78.76	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.22	0.00	0.00	62.50	7,757,072.00 *gal*
Application event totals		0.22	0.00	0.00	62.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	32.99	4,093,956.00 *gal*
Application event totals		0.12	0.00	0.00	32.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	32.99	4,093,956.00 *gal*
Application event totals		0.12	0.00	0.00	32.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	26.04	3,232,276.00 *gal*
Application event totals		0.09	0.00	0.00	26.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.17	0.00	0.00	46.88	5,817,967.00 *gal*
Application event totals		0.17	0.00	0.00	46.88	

Moran Ranch - 06/17/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.17	0.00	0.00	48.61	6,033,061.00 *gal*
Application event totals		0.17	0.00	0.00	48.61	

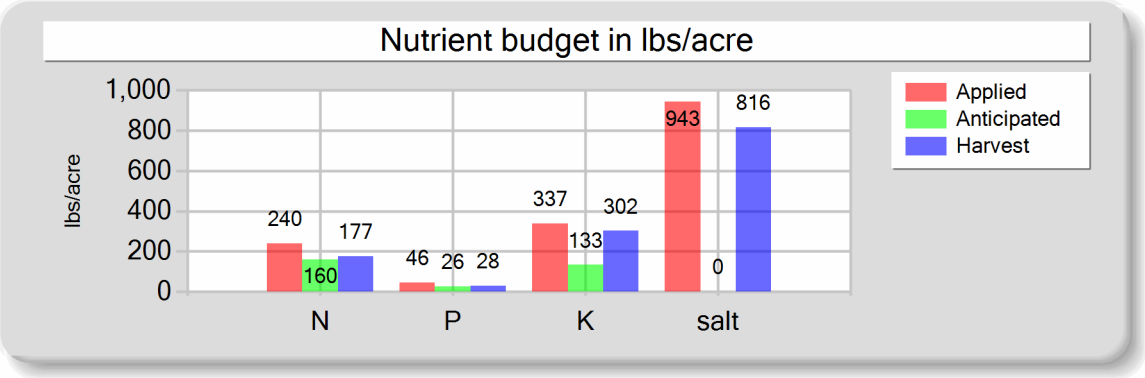
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	41.67	5,171,381.00 *gal*
Application event totals		0.15	0.00	0.00	41.67	

B. NUTRIENT BUDGET

Home Ranch - 10/05/2022: Oats, silage-soft dough

Field name: Home Ranch | Crop: Oats, silage-soft dough | Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	162.98	31.59	176.88	0.00
Process wastewater	72.06	14.47	160.62	875.51
Fresh water	0.24	0.00	0.00	67.25
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	239.95	46.06	337.50	942.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	176.78	28.34	301.61	816.44
Nutrient balance	63.16	17.72	35.89	126.32
Applied to removed ratio	1.36	1.63	1.12	1.15

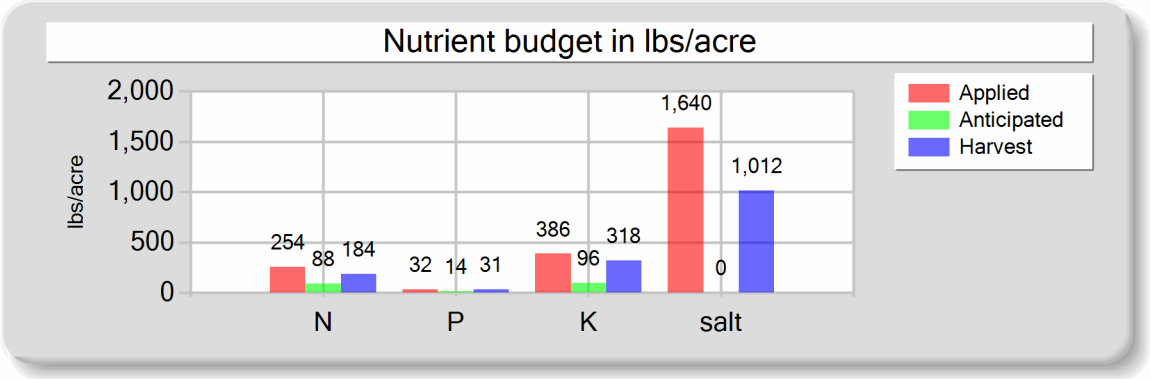
Fresh water applied	
8,058,204.00	*gallons*
296.76	*acre-inches*
10.60	*inches/acre*

Process wastewater applied	
1,500,000.00	*gallons*
55.24	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home Ranch - 06/20/2023: Sudangrass, silage

Field name: Home Ranch | Crop: Sudangrass, silage | Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	60.84	10.76	51.71	0.00
Process wastewater	182.36	21.47	334.49	1,353.50
Fresh water	1.02	0.00	0.00	286.55
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	253.56	32.23	386.20	1,640.05
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	183.56	30.50	318.25	1,012.04
Nutrient balance	70.00	1.73	67.94	628.01
Applied to removed ratio	1.38	1.06	1.21	1.62

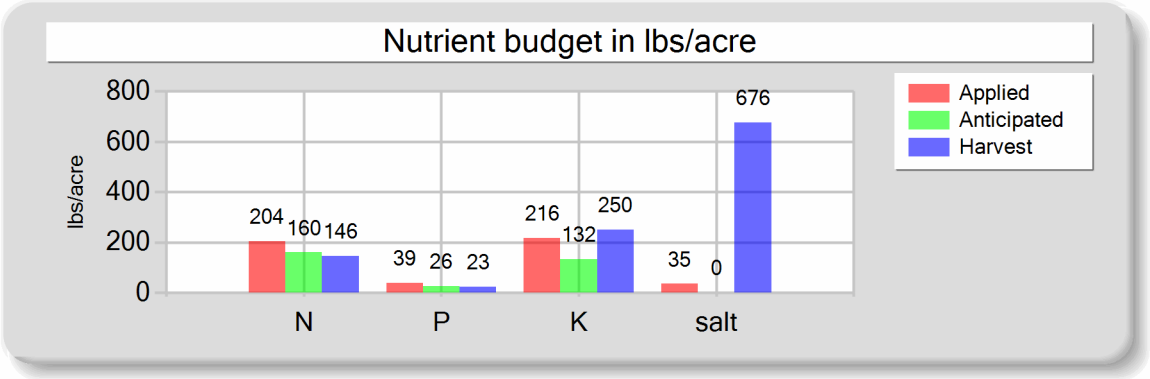
Fresh water applied	
34,338,128.00	*gallons*
1,264.56	*acre-inches*
45.16	*inches/acre*

Process wastewater applied	
870,000.00	*gallons*
32.04	*acre-inches*
1.14	*inches/acre*

Total harvests for the crop	
2	*harvests*

Moran Ranch - 10/05/2022: Oats, hay

Field name: Moran Ranch Crop: Oats, hay Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.33	38.63	216.32	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.12	0.00	0.00	34.72
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	204.12	38.63	216.32	34.72
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	146.36	23.46	249.70	675.93
Nutrient balance	57.76	15.17	-33.38	-641.21
Applied to removed ratio	1.39	1.65	0.87	0.05

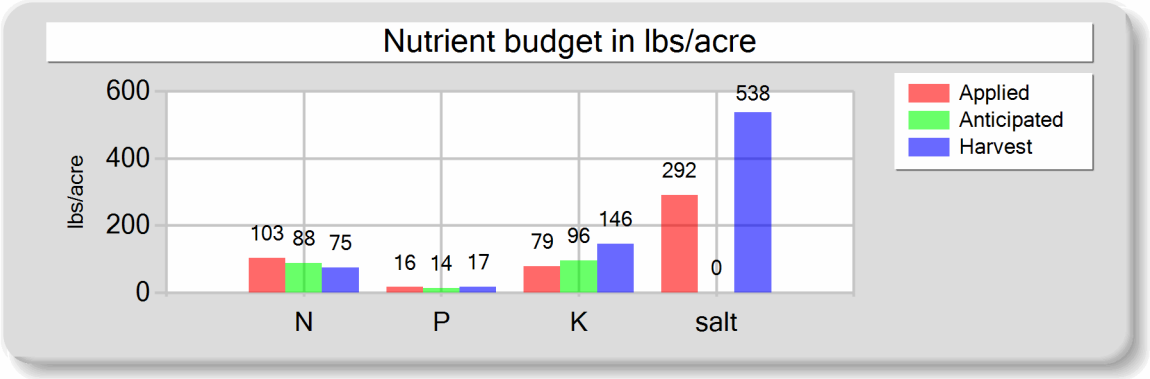
Fresh water applied	
4,309,702.00	*gallons*
158.71	*acre-inches*
5.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Moran Ranch - 06/17/2023: Sudangrass, silage

Field name: Moran Ranch Crop: Sudangrass, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	92.66	16.39	78.76	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.04	0.00	0.00	291.67
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	103.03	16.39	78.76	291.67
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	74.86	17.19	146.30	537.84
Nutrient balance	28.17	-0.79	-67.54	-246.17
Applied to removed ratio	1.38	0.95	0.54	0.54

Fresh water applied	
36,199,669.00	*gallons*
1,333.11	*acre-inches*
45.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure - 2022

Sample and source description: Separator Manure - 2022

Sample date: 12/26/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 26.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,800.00	5,000.00	28,000.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 05/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,000.00	4,600.00	22,100.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,600.00	7,000.00	26,600.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	193.00	129.00	0.00	1.00	39.60	436.00								4,160.00	2,160
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	135.00	84.90	0.00	0.40	26.70	299.00								3,040.00	1,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	138.00	92.80	0.00	0.30	31.50	273.00								2,890.00	1,640
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	702.00	427.00	0.00	1.30	82.80	1,290.00								9,140.00	5,220
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,140.00	428.00	0.00	0.10	322.00	2,020.00								12,000.00	6,550
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

C. FRESH WATER ANALYSES

Canal

MID Canal

Sample description: MID Canal

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.10	0.00								32.20	28
DL		0.01	0.01								0.01	1

Domestic Well

DW - Barn

Sample description: DW - Barn

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	3.90								282.00	
DL		0.01	0.01								0.01	

Domestic Well

DW - House

Sample description: DW - House

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	3.90								282.00	
DL		0.01	0.01								0.01	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Home Ranch - 10/05/2022: Oats, silage-soft dough

Oat Silage

Sample and source description: Oat Silage

Sample date: 05/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	4,200.00	44,700.00		12.10
DL	100.00	100.00	100.00		1.00

Home Ranch - 06/20/2023: Sudangrass, silage

Home Ranch - 06/20/2023: Sudangrass, silage

Dairy Sudan

Sample and source description: Dairy Sudan

Sample date: 08/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,600.00	4,600.00	56,800.00		17.40
DL	0.01	0.01	0.01		1.00

Sudan 2nd Cut

Sample and source description: Sudan 2nd Cut

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,600.00	5,700.00	48,300.00		16.20
DL	100.00	100.00	100.00		1.00

Moran Ranch - 10/05/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	4,200.00	44,700.00		12.10
DL	100.00	100.00	100.00		1.00

Moran Ranch - 06/17/2023: Sudangrass, silage

Moran Ranch - 06/17/2023: Sudangrass, silage

Moran Sudan

Sample and source description: Moran Sudan

Sample date: 08/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	4,800.00	41,000.00		16.10
DL	0.01	0.01	0.01		1.00

Sudan 2nd Cut

Sample and source description: Sudan 2nd Cut

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,600.00	5,700.00	48,300.00		16.20
DL	100.00	100.00	100.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

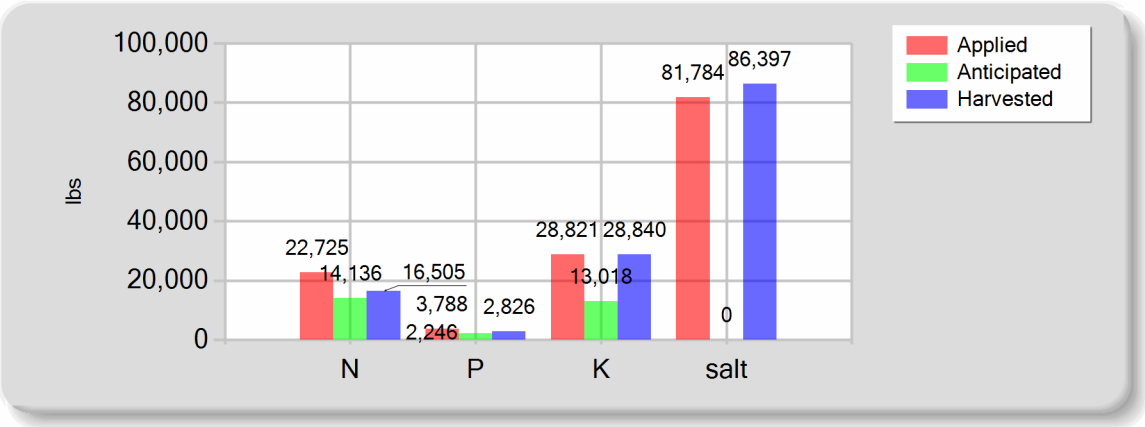
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

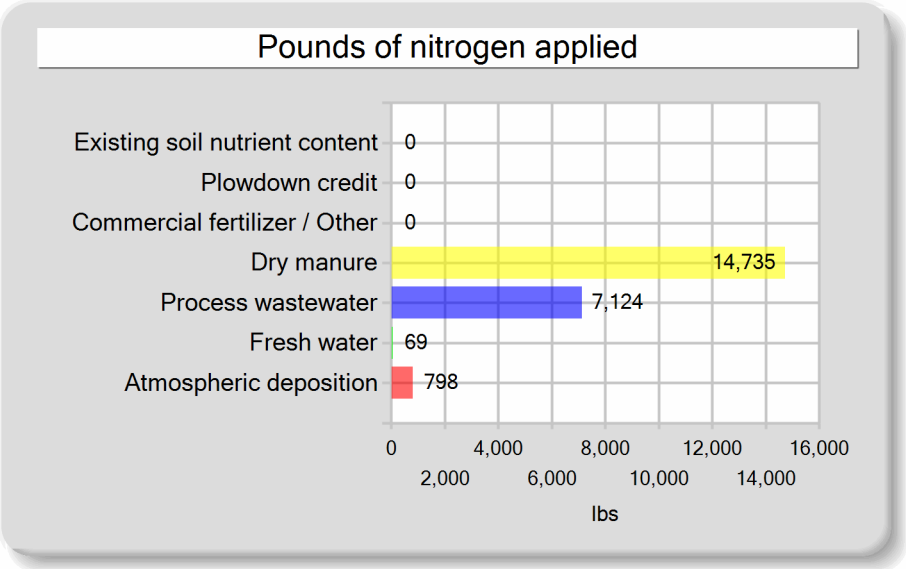
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

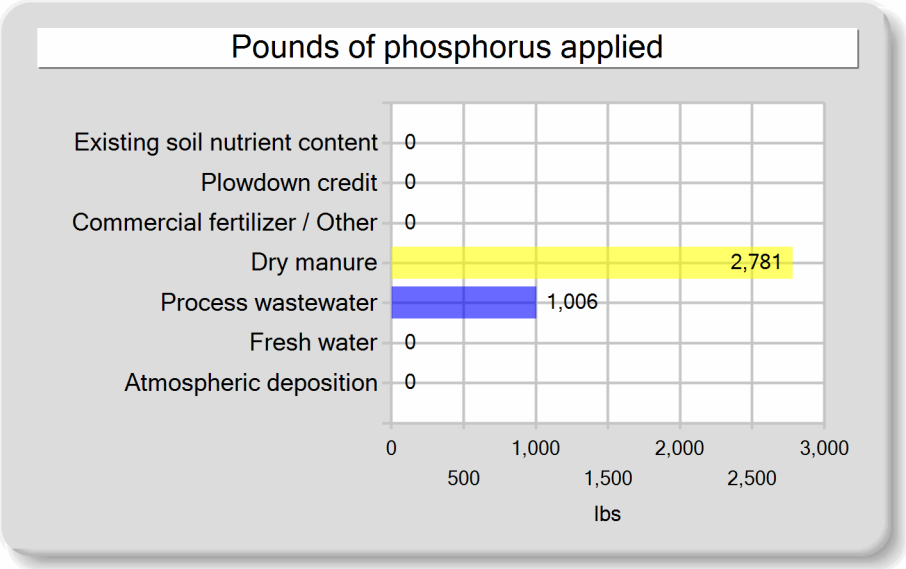
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	14,734.56	2,781.44	14,958.01	0.00
Process wastewater	7,123.68	1,006.41	13,862.88	62,412.26
Fresh water	69.18	0.00	0.00	19,371.75
Atmospheric deposition	798.00	0.00	0.00	0.00
Total nutrients applied	22,725.43	3,787.85	28,820.89	81,784.00
Anticipated crop nutrient removal	14,136.00	2,246.00	13,018.40	0.00
Actual crop nutrient removal	16,504.83	2,826.35	28,840.17	86,396.77
Nutrient balance	6,220.60	961.50	-19.28	-4,612.77
Applied to removed ratio	1.38	1.34	1.00	0.95

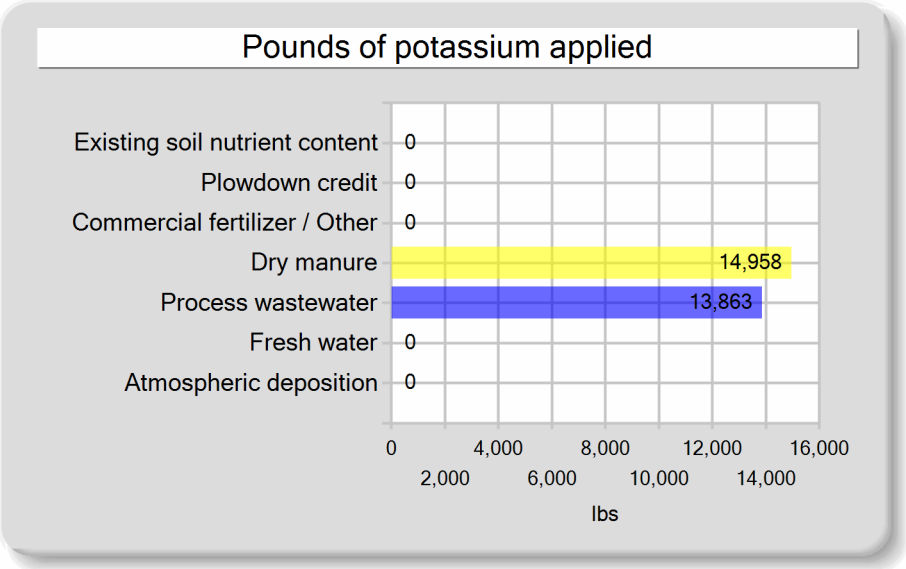
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

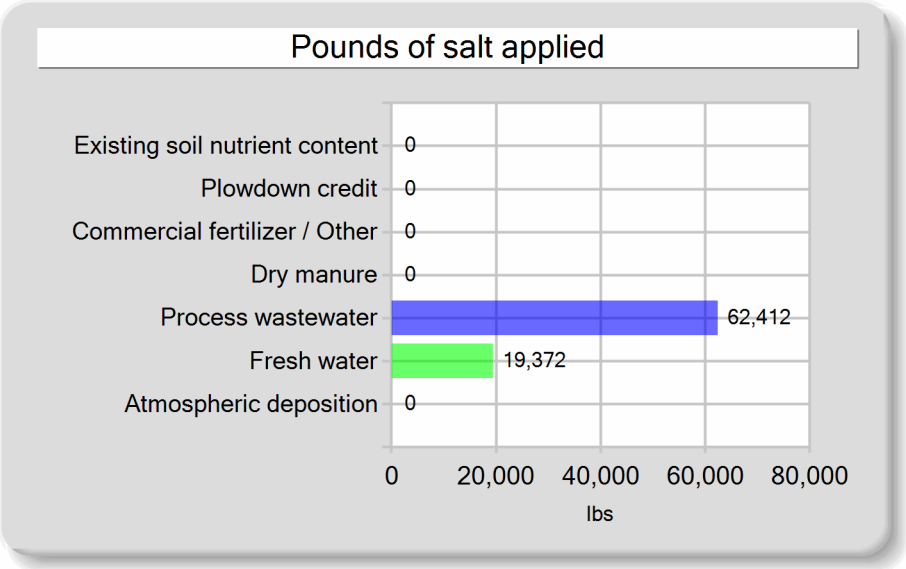


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Eulalia Borges	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Antero & Eulalia Borges	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7/1/24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Antero Borges Dairy
2260 Hull Road
Atwater CA 95301
00-0021389 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0884
Sampled By: M Dekeyrel
Requested By: Antero & Eulalia Borges
Submitted Date: 12/13/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - Barn	12/12/23	9:00	282	0.28	3.9	ND		7.8
2	DW - House	12/12/23	9:10	282	0.28	3.9	ND		7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Antero Borges Dairy
2260 Hull Road
Atwater CA 95301
00-0021389 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0884
Sampled By: M Dekeyrel
Requested By: Antero & Eulalia Borges
Submitted Date: 12/13/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - Barn	12/12/23	9:00	282	0.28	**3.9**	10.65	0.89
2	DW - House	12/12/23	9:10	282	0.28	**3.9**	10.65	0.89

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Antero Borges Dairy
2260 Hull Road
Atwater, CA 95301

Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0884-01	DW - Barn	Ag Water	M Dekeyrel		12/12/2023 9:00
23L0884-02	DW - House	Ag Water	M Dekeyrel		12/12/2023 9:10

Default Cooler Temperature on Receipt °C: 0.0
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Antero Borges Dairy
2260 Hull Road
Atwater, CA 95301

Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Sample Results

Sample: DW - Barn
23L0884-01 (Water)

Sampled: 12/12/2023 9:00
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		12/18/23 12:28	SM 2510 B		BEL0677
Electrical Conductivity umhos	**282**	umhos/cm	10.0	1		12/18/23 12:28	SM 2510 B		BEL0677
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 09:00	Field		BEL0567
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	12/17/23 15:22	EPA 300.0		BEL0663
Temperature	**25.0**	units	0.0	1		12/18/23 12:28	SM 4500-H+	H	BEL0677
pH	**7.8**	units	1.0	1		12/18/23 12:28	SM 4500-H+	H	BEL0677

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DELLAVALLE™
LABORATORY INC

Antero Borges Dairy
2260 Hull Road
Atwater, CA 95301

Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Sample Results
(Continued)

Sample: DW - House
23L0884-02 (Water)

Sampled: 12/12/2023 9:10
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		12/18/23 12:37	SM 2510 B		BEL0677
Electrical Conductivity umhos	**282**	umhos/cm	10.0	1		12/18/23 12:37	SM 2510 B		BEL0677
Ammonia (as N)	ND	mg/L	0.00	1		12/12/23 09:10	Field		BEL0567
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	12/17/23 15:43	EPA 300.0		BEL0663
Temperature	**25.0**	units	0.0	1		12/18/23 12:37	SM 4500-H+	H	BEL0677
pH	**7.9**	units	1.0	1		12/18/23 12:37	SM 4500-H+	H	BEL0677

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DELLAVALLE™
LABORATORY INC

Antero Borges Dairy
2260 Hull Road
Atwater, CA 95301

Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0663										
Blank (BEL0663-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0663-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0663-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0663-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0663-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0663-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		91.5	90-110		
LCS (BEL0663-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		93.0	90-110		
LCS (BEL0663-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		91.6	90-110		
LCS (BEL0663-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		91.7	90-110		
Duplicate (BEL0663-DUP1)		**Source: 23L0852-01**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	0.06		0.1	mg/L		0.06			0.00	10
Duplicate (BEL0663-DUP2)		**Source: 23L0860-01**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.9		0.1	mg/L		1.8			1.56	10
Duplicate (BEL0663-DUP3)		**Source: 23L0860-09**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.9			1.84	10
Duplicate (BEL0663-DUP4)		**Source: 23L0884-01**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.0		0.1	mg/L		3.9			1.37	10
Matrix Spike (BEL0663-MS1)		**Source: 23L0852-01**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.06	97.4	90-110		
Matrix Spike (BEL0663-MS2)		**Source: 23L0860-01**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.8	99.4	90-110		

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DELLAVALLE™ LABORATORY INC

Antero Borges Dairy
2260 Hull Road
Atwater, CA 95301

Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0663 (Continued)										
Matrix Spike (BEL0663-MS3)			**Source: 23L0860-09**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.9	98.6	90-110		
Matrix Spike (BEL0663-MS4)			**Source: 23L0884-01**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	8.9		0.1	mg/L	5.000	3.9	98.8	90-110		
Reference (BEL0663-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.6	90-110		
Reference (BEL0663-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.0	90-110		
Reference (BEL0663-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		96.5	90-110		
Reference (BEL0663-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.8	90-110		
Reference (BEL0663-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.0	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0677										
Blank (BEL0677-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0677-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.8		1.0	units						
Blank (BEL0677-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0677-DUP1)		**Source: 23L0885-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.60		0.01	mmhos/cm		0.60			0.268	10
pH	7.9		1.0	units		7.9			0.380	10
Electrical Conductivity umhos	597		10.0	umhos/cm		599			0.268	10
Duplicate (BEL0677-DUP2)		**Source: 23L0892-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.89		0.01	mmhos/cm		0.88			0.248	10
Electrical Conductivity umhos	887		10.0	umhos/cm		885			0.248	10
pH	7.8		1.0	units		7.8			0.384	10
Reference (BEL0677-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	429			umhos/cm	426.0		101	90-110		
Reference (BEL0677-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEL0677-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0677-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0677-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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LABORATORY INC

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Account# 00-0021389
Account Manager: Ben Nydam
Submitted By: Antero & Eulalia Borges

Received: 12/13/2023 15:35
Reported: 12/21/2023 08:54

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0677 (Continued)										
Reference (BEL0677-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0677-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0677-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0677-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21389 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: **Borges, Antero Dairy**
Address: 2260 Hull Road
City, State, Zip: Atwater, CA 95301
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Antero & Eulalia Borges - (209) 358-0680

Ranch _____

Date sampled: 12/12/23

Sampled by: mDekeuel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-Barn	Sampled From:	12/12/23	9AM	Ø	0.0
2. DW-House	Sampled From:	12/12/23	910AM	Ø	0.4
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MD	OASINC	12/12/23 9AM	12/13/23 11:20A
Second	JLF	DLV	12-13-23 11:00	12-13-23 3:01
Third				
Fourth	[signature]	DLL	12/13/23 15:35	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping

$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1								
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

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DLI SEND OUT